

## Beyond real solutions

$$x^2 = -9$$

**Why does this equation have no real solution?**

## The imaginary unit

The imaginary unit is defined by its square.

$$i^2 = -1$$

**What value does  $(3i)^2$  have?**

## A complex number

A complex number has the form  $a + bi$ , where  $a$  and  $b$  are real.

$$a + bi$$

**How are real numbers included among complex numbers?**

## Identify the parts

$$-4 + 7i$$

**What are this number's real part and imaginary part?**

## A negative radicand

Use the principal complex square root.

$$\sqrt{-50}$$

**How can we write this radical in standard complex form?**

## Two square roots

$$z^2 = -25$$

**What are both complex solutions?**

## Add complex numbers

$$(3 - 2i) + (5 + 7i)$$

**What is the sum in standard form?**

## Subtract complex numbers

$$(2 + 6i) - (5 - 3i)$$

**What is the difference in standard form?**

## Multiply complex numbers

$$(2 + i)(3 - 4i)$$

**What is the product in standard form?**

## Powers of $i$

$$i^7$$

**How can we simplify this power?**

## Complex conjugates

The conjugate of  $a + bi$  is  $a - bi$ .

$$4 + 3i$$

**What happens when this number is multiplied by its conjugate?**

## Divide by a complex number

$$\frac{5}{2-i}$$

**What is the quotient in standard form?**

## Complex roots of a quadratic

$$(x - 2)^2 = -12$$

**What are the complex solutions in exact form?**

## A quadratic factor

$$(x - i)(x + i)$$

**What real polynomial does this product produce?**